

Rates and Unit Rates

Flagship

Lesson 4-1-flagship

Name: _____

Date: _____

Class: _____

ARCADE BUILDER MISSION

Best Booth in the Arcade

You are the new operations manager at NeonPlex Arcade. Two ticket booths are competing for players – Booth A charges \$3 for 5 games, Booth B charges \$5 for 8 games. Players want the best deal, and your job is to crown the winning booth by mastering unit rates.

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

<i>\$12 for 4 games is a rate</i> Rate A ratio comparing two amounts with different units, like miles per hour.	<i>\$3 per 1 game</i> Unit Rate A rate for just 1 of something, like cost for 1 item.
<i>60 miles per hour means 60 miles for every 1 hour</i> Per For each one. Example: 5 dollars per book.	<i>5 to 3 or 5:3</i> Ratio A way to compare two amounts.
<i>Pack A: \$0.50 per pencil vs Pack B: \$0.40 per pencil — Pack B is the better buy</i> Better Buy The choice that costs less for each item.	<i>$\\$12 \div 4 \text{ games} = \\3 per game</i> Divide Splitting a total into equal groups.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find a unit rate to compare prices and decide the better buy.

1 Notice

You're comparing ticket prices at different game booths in the arcade.

2 Key idea

I can find a unit rate to compare prices and decide the better buy.

3 Words I'll use

Rate

Unit Rate

Per

Ratio

Better Buy

Divide



Think About It

- What two quantities are being compared at each booth?
- Which booth seems like a better deal at first glance?
- How can you compare prices when the number of games is different?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: A booth charges \$4.50 for 9 games. What is the unit rate?

- A. \$0.50 per game
- B. \$0.45 per game
- C. \$2.00 per game
- D. \$4.50 per game

Step 1

$$\$4.50 \div 9 = \$0.50 \text{ per game.}$$



Answer: A. \$0.50 per game

 We do — together

Solve this one with your class using the same steps.

Problem: A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

- A. 30 miles per gallon
- B. 6 miles per gallon
- C. 180 miles per gallon
- D. 36 miles per gallon

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A pack of 8 markers costs \$6.00. What is the cost per marker?

- A. \$0.75
- B. \$0.80
- C. \$1.33
- D. \$0.60

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Booth A charges \$3 for 5 games and Booth B charges \$5 for 8 games. Can you tell which is the better deal just by looking at the signs? Why or why not?

Level 1 support

Start here: You can't compare fairly until you find the cost per 1 game (the unit rate) for each booth.

 **Need a hint?**

Sentence starters (optional)

You ____ tell just by looking because the number of ____ is different.

____ se puede saber solo mirando porque la cantidad de ____ es diferente.

To compare fairly I need the cost per ____ game.

Para comparar de forma justa necesito el costo por ____ juego.

Word bank: rate unit rate per better buy compare

Listen for: Students recognize the booths offer different numbers of games, so a fair comparison needs the cost per 1 game (unit rate).

Level 2

Some players pick Booth B because \$5 buys more games. Why is that reasoning not enough to know the better deal? Justify using cost per game.

- Buying more games does not always mean a better deal because ____.
- The real test is the ____, since ____.

EXPLORE

How did you find the cost per game for each booth, and which booth wins?

Level 1 support

Start here: Divide the price by the number of games to get the unit rate, then compare the two rates.

 **Need a hint?**

Sentence starters (optional)

Booth A costs ____ per game because 3 divided by 5 equals ____.

El puesto A cuesta ____ por juego porque 3 entre 5 es ____.

The better buy is Booth ____ because its ____ is lower.

La mejor compra es el puesto ____ porque su ____ es más baja.

Word bank: **rate** **unit rate** **per** **better buy** **divide**

Listen for: Students compute \$0.60/game for Booth A and \$0.625/game for Booth B, then choose Booth A because the lower unit rate is the better buy.

Level 2

If a player only wants 5 games total, does the better unit rate always give the cheaper total? Explain when it might not.

- The lower unit rate gives the cheaper total only when ____.
- It might not be cheaper if ____.

CONNECT

At the snack bar, how would you use a unit rate to find the best popcorn deal among different sizes?

Level 1 support

Start here: Find the price per ounce (or per cup) for each size, then pick the lowest unit rate.



Need a hint?

Sentence starters (optional)

I would find the price per ____ for each popcorn size.

Hallaría el precio por ____ para cada tamaño de palomitas.

The best deal is the one with the ____ unit rate.

La mejor oferta es la que tiene la ____ tasa unitaria.

Word bank: **rate** **unit rate** **per** **better buy** **compare**

Listen for: Students describe dividing price by size to get a per-unit cost and choosing the lowest unit rate as the best deal.

Level 2

Why might the biggest popcorn not always be the best value? Use the idea of unit rate to explain.

- The biggest size is not always the best value because ____.
- The unit rate shows the real value because ____.

Write About the Math

The Writing Revolution

I can explain my choice using the words rate, unit rate, per, and better buy.

1. Kernel Sentence subject + verb

Model: Rate is a ratio comparing two amounts with different units, like miles per hour.

Tasa es una razón que compara dos cantidades con unidades distintas, como millas por hora.

Write a kernel sentence about rate. Use a subject and a verb.

Escribe una oración base sobre tasa. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Rate matters in math

Tasa importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Rate matters in math because ____.

Tasa importa en matemáticas porque ____.

but
pero

Rate matters in math, but ____.

Tasa importa en matemáticas, pero ____.

so
entonces

Rate matters in math, so ____.

Tasa importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about rate.
Di un hecho verdadero sobre rate.

Rate ____.

Question
Pregunta

Ask a question about rate.
Haz una pregunta sobre rate.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about rate.
Muestra entusiasmo sobre rate.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with rate.
Dile a un compañero qué hacer con rate.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

You ____ **tell just by looking because the number of** ____ **is different.**

____ se puede saber solo mirando porque la cantidad de ____ es diferente.

To compare fairly I need the cost per ____ **game.**

Para comparar de forma justa necesito el costo por ____ juego.

Booth A costs ____ **per game because 3 divided by 5 equals** ____.

El puesto A cuesta ____ por juego porque 3 entre 5 es ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. Which token pack from the table above is the best deal?

- A. Pack A
- B. Pack B
- C. Pack C
- D. They are all the same

Show your work:

2. Runner A completes 4 laps in 12 minutes. Runner B completes 5 laps in 14 minutes. Who runs more laps per minute?

- A. Runner B
- B. Runner A
- C. Same rate
- D. Cannot determine

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

An arcade sells tokens in three packs: 25 tokens for \$5, 60 tokens for \$10.80, and 100 tokens for \$19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Sentence starter: The student is ____ because the best deal is the ____ pack with a unit rate of \$____ per token.

Show your work:

Reflect — Exit Ticket

A store sells 6 pencils for \$1.50 and 10 pencils for \$2.80. Which is the better buy?

- A. 6 for \$1.50 — \$0.25 each
- B. 10 for \$2.80 — \$0.28 each
- C. They cost the same per pencil
- D. Not enough information

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 30 miles per gallon — $180 \text{ miles} \div 6 \text{ gallons} = 30 \text{ miles per gallon}$.
2. **You Try (notes):** A. $\$0.75$ — $\$6.00 \div 8 = \0.75 per marker .
3. **Try It 1:** B. Pack B — *Pack B has a unit rate of \$0.20 per token, which is the lowest cost per token.*
4. **Try It 2:** A. Runner B — *Runner A: $4 \div 12 \approx 0.33 \text{ laps/min}$. Runner B: $5 \div 14 \approx 0.36 \text{ laps/min}$. Runner B runs slightly more laps per minute.*
5. **Exit Ticket:** A. 6 for \$1.50 — $\$0.25$ each — $\$1.50 \div 6 = \0.25 per pencil . $\$2.80 \div 10 = \0.28 per pencil . $\$0.25 < \0.28 , so 6 for \$1.50 is the better buy.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rate is a ratio comparing two amounts with different units, like miles per hour.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).